

REPORT
OF
AICTE–EduSkills Virtual Internship Report: MERN Full Stack
Development with Project

SUBMITTED TO
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B. Tech CSE



BY
Name of Student: MANISH KUMAR JAISWAL
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Batch: 2026-27

CANDIDATE'S DECLARATION

I, Manish Kumar Jaiswal, a student of B.Tech (Computer Science and Engineering) at IILM University, Greater Noida, do hereby declare that the internship report titled “AICTE–EduSkills Virtual Internship Report: MERN Full Stack Development with Project” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references used have been duly acknowledged, and I have not taken as such any material from any other source without citation. I affirm that this report is my own original work and has not been submitted earlier to any other institute for the award of any degree. I shall be solely responsible for any kind of copyright violation in this regard.

Signature: Manish

Student Name: Manish Kumar Jaiswal

Roll No.: 2410030307

Date: 23-08-2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to AICTE and EduSkills for providing me the opportunity to undertake this virtual internship in MERN Full Stack Development with Project. I am thankful to my industry mentor, Ms. Sagarabala Behera, EduSkills Academy, for her valuable guidance, continuous support, and technical insights throughout the internship period.

I would also like to thank Mr. Hilal, School of Computer Science and Engineering, IILM University, for the encouragement and academic guidance extended during this internship. I am also grateful to the Department of Computer Science and Engineering for its constant support.

Finally, I extend my heartfelt thanks to my family and friends for their encouragement throughout this internship.

Signature: Manish

Student Name: Manish Kumar Jaiswal

Roll No.: 2410030307

INTERNSHIP COMPLETION CERTIFICATE

The following is reproduced from the official AICTE–EduSkills internship completion certificate issued to the candidate:

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Mr. Manish Kumar Jaiswal, a student of IILM University, Greater Noida, has successfully completed an 8-Week (June 2026 – July 2026) AICTE – EduSkills Virtual Internship Program in MERN Full Stack Development With Project. During the internship, he worked on the project titled “JSON Database Simulator Project.” His performance during the internship was found to be satisfactory, and he demonstrated good technical and professional skills.

Mentor: Sagarabala Behera, EduSkills Academy

AICTE Student ID: STU699acefa8d6101771753210

APAAR ID: 021111601957

Certificate Date: 23-08-2026

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5. PROJECT DESCRIPTION

5.1 Introduction

Industry-oriented internships play an important role in bridging the gap between classroom learning and real-world software development practice. As part of the B.Tech (Computer Science and Engineering) curriculum, I undertook an 8-week AICTE–EduSkills Virtual Internship in the domain of MERN Full Stack Development With Project. The internship was conducted entirely online through the EduSkills Learning Management System and was designed to provide practical exposure to modern web-development technologies alongside the theoretical concepts learnt in the classroom.

During the internship, I studied and practised core concepts of full-stack JavaScript development, including React, Redux Toolkit, React Router, TailwindCSS, Node.js, Express.js, MongoDB, Mongoose, RESTful APIs, JWT-based authentication, and basic Git and deployment workflows. Alongside these topics, hands-on project work was undertaken, including an E-commerce Platform and the JSON Database Simulator Project. This report focuses primarily on the JSON Database Simulator Project, as it represents the most detailed piece of project work completed and documented during the internship, while also summarising the broader internship experience.

5.2 Organization Profile

Organization Name: EduSkills Academy

Domain: EdTech & Skill Development

Services Offered: Virtual Internships in collaboration with AICTE

Vision: To empower students with industry-relevant skills.

5.3 Problem Statement

Most academic exercises expose students to database concepts only through the use of ready-made database management systems, leaving the underlying mechanics of data storage and retrieval largely abstracted away. The JSON Database Simulator Project was undertaken to address this gap by building, from first principles, a Node.js

application that replicates the core functionalities of a database using JSON as the underlying data structure. The simulator needed to support structuring of data into collections, performing Create, Read, Update, and Delete (CRUD) operations on records, retrieving records through filtering and sorting, and persisting the current state of the database to a JSON file.

5.4 Project Objectives

- To design and implement a JSON-based database simulator using Node.js.
- To understand how database-like data structures and operations can be represented programmatically.
- To implement Create, Read, Update, and Delete (CRUD) operations on structured data.
- To implement filtering and sorting functionality for data retrieval.
- To gain hands-on experience with Node.js file I/O for data persistence.
- To improve overall programming and problem-solving skills.
- To build a foundational understanding of backend and data-management concepts.

5.5 Scope of the Project

The JSON Database Simulator is intended as a learning-oriented tool for understanding data-management principles rather than as a production database system. Within this scope, it supports defining collections, inserting and modifying records, deleting records, and retrieving data through filtering and sorting, with the resulting state exportable as a JSON file. The project can potentially be extended in the future with advanced querying (such as range-based or text search), support for multiple named collections, a command-line interface for interaction, and basic indexing to improve search performance. These extensions were identified as future scope and were not implemented as part of the current internship project.

5.6 Technologies and Tools Used

Table 5.1: Technologies Covered During the Internship

Category	Technologies
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Frontend	React, Redux Toolkit, React Router, TailwindCSS
Backend	Node.js, Express.js
Database	MongoDB, Mongoose
APIs & Security	RESTful APIs, JWT Authentication
Tools	Git and basic deployment workflow

Table 5.2: Technologies Used in the JSON Database Simulator Project

Category	Technologies / Concepts
Language	JavaScript (Node.js)
Core Module	Node.js fs module (File I/O)
Data Format	JSON
Concepts	CRUD operations, filtering, sorting

5.7 System Architecture

The JSON Database Simulator follows a simple, linear flow of data from user input through to persistent storage, as shown in Figure 5.1.

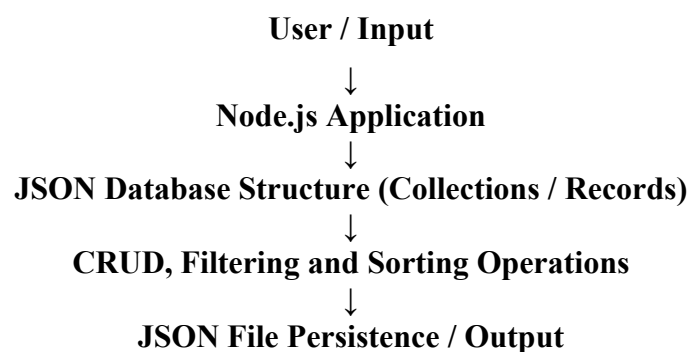


Figure 5.1: Conceptual Architecture of the JSON Database Simulator

5.8 Methodology

The project was developed in the following logical stages:

- Stage 1: Defining the JSON database structure using nested JSON objects to represent collections and records.
- Stage 2: Implementing functionality to create and structure new collections.
- Stage 3: Implementing insertion of new records into the appropriate collections.

- Stage 4: Implementing update operations to modify existing records based on identifying criteria.
- Stage 5: Implementing deletion of records while preserving the integrity of the remaining data.
- Stage 6: Implementing filtering to retrieve records matching specified conditions.
- Stage 7: Implementing sorting to organise retrieved records by one or more attributes.
- Stage 8: Implementing JSON file persistence to export the current state of the database.
- Stage 9: Testing and verifying each operation for correctness.

Implementation Overview

The simulator was implemented around five core operations. The database structure was first created as a nested JSON object representing collections and their records. Insertion logic appended new, schema-conformant records into the appropriate collection. Update logic located a target record using identifying criteria (implemented with array methods such as `findIndex`) and modified its attributes in place, while delete logic removed a specific record without disturbing the rest of the collection (implemented using `filter`). Filtering and sorting were implemented using JavaScript's built-in array methods to retrieve subsets of data and arrange them by one or more attributes. Finally, the Node.js `fs` module was used to write the resulting database state to a JSON file, giving a persistent, human-readable snapshot of the data.

5.9 Expected Outcomes / Learning Outcomes

- Gained practical experience implementing core database operations (CRUD) using JavaScript and JSON.
- Developed a deeper understanding of data structuring and manipulation outside a relational-database context.
- Enhanced skills in file I/O operations for data persistence and retrieval.
- Improved proficiency in applying filtering and sorting logic to dynamic datasets.

- Acquired foundational knowledge of backend and data-management concepts.
- Gained broader exposure to full-stack technologies, including React, Node.js, Express.js and MongoDB, as covered during the internship.

Challenges encountered during the project, and how they were addressed, are summarised in Table 5.3.

Table 5.3: Challenges and Solutions

Challenge	Solution
Maintaining data integrity during update and delete operations	Used array methods such as filter and findIndex with careful validation to ensure operations targeted the correct records.
Efficiently filtering and sorting datasets	Used JavaScript's built-in filter and sort array methods, which are optimised for performance.
Handling file I/O safely during data export	Implemented error handling around file-write operations to manage issues such as permission errors gracefully.

Overall, the JSON Database Simulator project provided practical, hands-on experience in programming, backend development, and data management, and complemented the broader MERN stack concepts covered during the internship. It also reinforced the value of structured, stage-wise development and careful testing when building even a small-scale software system.

5.10 Certificates of Completion and Internship Proof

A copy of the official AICTE–EduSkills internship completion certificate is reproduced in Section 4 (Internship Completion Certificate) of this report. [Attach internship offer letter/email communication here, if required by the department.]

6. BIBLIOGRAPHY / REFERENCES

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2. MDN Web Docs – JavaScript Arrays. Available at: https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Array
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